WORKED SOLUTION

# STEP II 2000, Question 10

Mechanics • Connected Pulley Systems

**STEP II 2000, Question 10**

Read the question before you go on. It is in the STEP Support Programme's question database:

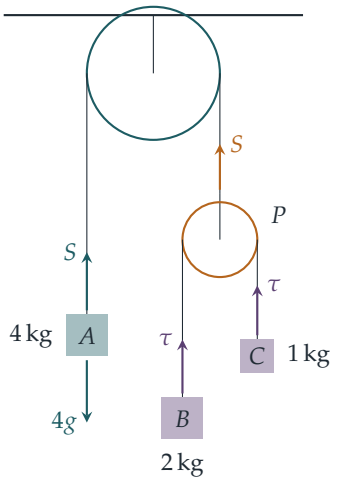
step.maths.org/questions/00-s2-q10

## The key idea

This is a double Atwood machine. While $A$ is held, the second pulley is fixed and $B$, $C$ form an ordinary Atwood machine. Once $A$ is let go, the second pulley moves too, so the clean way in is to measure the motion of $B$ and $C$ *relative to that pulley*, and to use the fact that a light pulley carries no net force: the string above it pulls with exactly twice the tension of the string below.

## Solution

We model the particles as point masses, the strings as light and inextensible, and both pulleys as smooth and light. Call the second pulley $P$. We take *downwards* as positive for every acceleration, write $S$ for the tension in the string through $A$ and $\tau$ for the tension in $BC$, and take $g$ in $\mathrm{m\,s^{-2}}$.




Because $P$ is light, the net force on it must be zero (otherwise it would have infinite

acceleration). It is pulled up by $S$ and down by $\tau$ on each side, so

$$S = 2\tau$$

at all times.

#### First phase: $A$ held

With $A$ held still, the string over the fixed pulley doesn't move, so $P$ is fixed. Then $B$ and $C$ are an ordinary Atwood machine: if $B$ accelerates down at $a$, then $C$ accelerates up at $a$. Newton's second law gives

$$2a = 2g - \tau, \qquad 1 \cdot a = \tau - g.$$

Adding, $3a = g$, so

$$a = \tfrac{1}{3}g, \qquad \tau = \tfrac{4}{3}g.$$

So $B$ accelerates downwards and $C$ upwards, each at $\frac{1}{3}g\ \mathrm{m\,s^{-2}}$.

At the end of this phase ($t = T$), $B$ is moving down and $C$ up, each with speed $\frac{1}{3}gT$, and $B$ has fallen $\frac{1}{2} \cdot \frac{1}{3}g\,T^2 = \frac{1}{6}gT^2$. Throughout, $A$ and $P$ are at rest.

#### Second phase: everything free

Let $A$ have acceleration $a_A$ (downwards). The string over the fixed pulley has constant length, so $P$ has acceleration $-a_A$: it goes up as fast as $A$ goes down.

> **Ask yourself** How do we describe the accelerations of $B$ and $C$ when the pulley they hang from is itself accelerating?

Relative to $P$, the string $BC$ behaves just as in a fixed Atwood machine: if $B$ moves down relative to $P$ with acceleration $r$, then $C$ moves up relative to $P$ with acceleration $r$. Adding $P$'s own acceleration,

$$a_B = -a_A + r, \qquad a_C = -a_A - r.$$

Now Newton's second law for each particle (downwards positive), using $S = 2\tau$:

$$\begin{aligned} A: &\quad 4a_A = 4g - 2\tau, \\ B: &\quad 2(-a_A + r) = 2g - \tau, \\ C: &\quad 1(-a_A - r) = g - \tau. \end{aligned}$$

From $A$, $\tau = 2g - 2a_A$. Putting this into $B$ gives $-2a_A + 2r = 2a_A$, so $r = 2a_A$. Putting both into $C$:

$$-3a_A = g - (2g - 2a_A) = -g + 2a_A \quad \Longrightarrow \quad a_A = \tfrac{1}{5}g.$$

Hence $r = \frac{2}{5}g$, $\tau = \frac{8}{5}g$, $S = \frac{16}{5}g$, and

$$a_A = \tfrac{1}{5}g \text{ (down)}, \qquad a_B = -\tfrac{1}{5}g + \tfrac{2}{5}g = \tfrac{1}{5}g \text{ (down)}, \qquad a_C = -\tfrac{1}{5}g - \tfrac{2}{5}g = -\tfrac{3}{5}g,$$

so $C$ accelerates *upwards* at $\frac{3}{5}g$ m s$^{-2}$.

These are constant, and at the moment $A$ is released nothing changes abruptly: $A$ and $P$ start from rest, and $B$, $C$ keep their speeds $\frac{1}{3}gT$.

#### Speed of $A$

In the first phase $B$ moved $\frac{1}{6}gT^2$, so it still has

$$0.6\,gT^2 - \tfrac{1}{6}gT^2 = \tfrac{3}{5}gT^2 - \tfrac{1}{6}gT^2 = \tfrac{13}{30}gT^2$$

to go. In the second phase $B$ starts with downward velocity $\frac{1}{3}gT$ and has downward acceleration $\frac{1}{5}g$, so its velocity stays positive and the distance it travels equals its displacement. If this takes time $t$, then $s = ut + \frac{1}{2}at^2$ gives

$$\tfrac{1}{3}gT\,t + \tfrac{1}{10}gt^2 = \tfrac{13}{30}gT^2.$$

Write $t = kT$ and multiply by $30/(gT^2)$:

$$3k^2 + 10k - 13 = 0 \quad\Longrightarrow\quad (k-1)(3k+13) = 0.$$

Since $t > 0$, $k = 1$: the second phase lasts exactly $T$ seconds. $A$ started this phase from rest with acceleration $\frac{1}{5}g$, so its speed is then

$$v_A = \tfrac{1}{5}g \cdot T = \tfrac{gT}{5} \text{ m s}^{-1}.$$

## Checks

- Each equation of motion holds with the values found: for $A$, $4 \cdot \frac{1}{5}g = 4g - \frac{16}{5}g$; for $C$, $-\frac{3}{5}g = g - \frac{8}{5}g$.
- An independent check: the strings force $y_C = \text{const} - 2y_A - y_B$ (downward positions), and energy methods with the two free coordinates $y_A$, $y_B$ give the same $a_A = a_B = \frac{1}{5}g$, $a_C = -\frac{3}{5}g$.
- Stepping the motion numerically with $g = 9.8$, $T = 1.7$, $B$ has travelled $0.6gT^2$ at $t = 3.400$ (that is, $2T$), when $A$'s speed is $3.332 = gT/5$.
- Total weight is $7g$ but the fixed pulley only takes $2S = \frac{32}{5}g$, less than $7g$, so the system's centre of mass accelerates downwards in phase two, as it should.

## Common slips

- Treating $P$ as fixed in the second phase and reusing $\pm\frac{1}{3}g$ for $B$ and $C$.
- Forgetting that $S = 2\tau$ for a light pulley, or giving the pulley a mass term.
- Taking the full $0.6gT^2$ as the second-phase distance, instead of subtracting the $\frac{1}{6}gT^2$ already covered.

## Beyond STEP

Choosing positions that automatically satisfy the string constraints, as in the second check, is the starting point of Lagrangian mechanics: write the kinetic and potential energy in terms of the free coordinates and the equations of motion follow, with no tensions in sight. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats constrained systems such as pulleys carefully and is a good next step.

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